

# **Lecture 6: Four Fundamental Subspaces**

MATH203 Applied Linear Algebra

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## Motivation: Connecting the Subspaces

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## Recall: The Preview from Lecture 5

At the end of Lecture 5, we discovered something beautiful:

### Theorem (Preview)

$$\text{Null}(A) \perp \text{Row}(A)$$

Every vector in  $\text{Null}(A)$  is orthogonal to every row of  $A$ .

**Why?** If  $\mathbf{x} \in \text{Null}(A)$ , then  $A\mathbf{x} = \mathbf{0}$ , which means:

$$\text{row}_i \cdot \mathbf{x} = 0 \quad \text{for all rows } i$$

**Today's question:** Is this an isolated phenomenon, or part of a bigger picture?

# What We'll Discover Today

For any matrix  $A$ , there are **four fundamental subspaces**:

| Subspace           | Lives in                  | Dimension |
|--------------------|---------------------------|-----------|
| $\text{Col}(A)$    | $\mathbb{R}^m$ (codomain) | $r$       |
| $\text{Null}(A)$   | $\mathbb{R}^n$ (domain)   | $n - r$   |
| $\text{Col}(A^T)$  | $\mathbb{R}^n$ (domain)   | $r$       |
| $\text{Null}(A^T)$ | $\mathbb{R}^m$ (codomain) | $m - r$   |

**Discovery:** They pair up into **orthogonal complements**!

# Roadmap for Today

**Strategy:** Build tools first, then reveal the full picture.

**Part 1:** How do subspaces behave under matrix products? ( $AB$ )

**Part 2:** What is transpose? (Flipping the ingredient table)

**Part 3:** The four fundamental subspaces and their orthogonality

**Part 4:** Dimension relationships

**Goal:** Understand the complete geometric structure of a matrix.

# Subspaces of Matrix Products

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# Setup: Two-Stage Production

Imagine a **two-stage coffee shop**:



- $B$ : raw materials  $\rightarrow$  semi-products (e.g., roasted beans, brewed tea)
- $A$ : semi-products  $\rightarrow$  final products (e.g., latte, bubble tea)
- $AB$ : raw materials  $\rightarrow$  final products (combined process)

**Question:** How do the column spaces and null spaces of  $A$ ,  $B$ , and  $AB$  relate?

## Concrete Example: Two Matrices

Let's use actual matrices:

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad (3 \times 3)$$

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (2 \times 3)$$

**Product**  $AB$ :

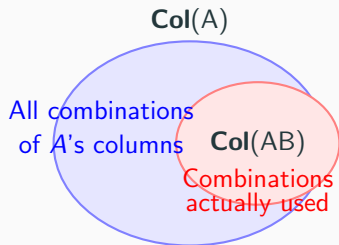
$$AB = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

We'll investigate: What are  $\text{Col}(A)$ ,  $\text{Col}(B)$ ,  $\text{Col}(AB)$ ?



# Visual: Column Space Containment

**Key observation:** Every column of  $AB$  is built from columns of  $A$ .



**Why?** Column  $j$  of  $AB$  equals  $A$  times column  $j$  of  $B$ .

So every column of  $AB$  is a **linear combination of  $A$ 's columns**.

# Theorem 1: Column Space of Product

## Theorem

$$\text{Col}(AB) \subseteq \text{Col}(A)$$

The column space of  $AB$  is always contained in the column space of  $A$ .

## Proof:

Write  $B$  with columns  $\mathbf{b}_1, \dots, \mathbf{b}_n$ :

$$B = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Then  $AB$  has columns:

$$AB = \begin{pmatrix} | & | & \cdots & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Each  $A\mathbf{b}_j$  is a linear combination of  $A$ 's columns.



# Why This Makes Sense: Column View of Matrix Product

Recall from Lecture 1: column  $j$  of  $AB$  equals  $A \cdot$  (column  $j$  of  $B$ ).

**Example:** If  $\mathbf{b}_j = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$  and  $A$  has columns  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ :

$$A\mathbf{b}_j = 2\mathbf{a}_1 - 1\mathbf{a}_2 + 3\mathbf{a}_3$$

This is a **recipe** using  $A$ 's columns as ingredients!

**Conclusion:** Every column of  $AB$  is cooked from  $A$ 's columns.

Therefore  $\text{Col}(AB) \subseteq \text{Col}(A)$ .

## Verify with Our Example

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

Column space of  $AB$ :

$$\text{Col}(AB) = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\}$$

Notice:  $\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  (dependent!)

So actually:  $\text{Col}(AB) = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$

Column space of  $A$ :

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$$

Indeed:  $\text{Col}(AB) \subset \text{Col}(A)$ .  $\checkmark$  (Equality in this case!)

## Theorem 2: Null Space Inclusion

### Theorem

$$\text{Null}(B) \subseteq \text{Null}(AB)$$

The null space of  $B$  is always contained in the null space of  $AB$ .

### Proof:

If  $\mathbf{x} \in \text{Null}(B)$ , then  $B\mathbf{x} = \mathbf{0}$ .

Therefore:

$$(AB)\mathbf{x} = A(B\mathbf{x}) = A(\mathbf{0}) = \mathbf{0}$$

So  $\mathbf{x} \in \text{Null}(AB)$ .



## Why This Makes Sense: Linear Dependencies Persist

Recall from Lecture 5:  $\mathbf{x} \in \text{Null}(B)$  records a **linear dependence** among columns of  $B$ .

**Example:** If  $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \in \text{Null}(B)$ , this says:

$$2\mathbf{b}_1 - 1\mathbf{b}_2 + 0\mathbf{b}_3 = \mathbf{0}$$

**What happens when we apply  $A$  to both sides?**

$$A(2\mathbf{b}_1 - 1\mathbf{b}_2) = 2(A\mathbf{b}_1) - 1(A\mathbf{b}_2) = 2(AB)_1 - 1(AB)_2 = \mathbf{0}$$

**Conclusion:** The **same linear dependence** holds for columns of  $AB$ !

Applying  $A$  preserves all existing linear relations.

## Verify with Our Example

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

**Find Null( $B$ ):**

By inspection, column 3 of  $B$  equals  $2 \times$  column 1 plus  $1 \times$  column 2:

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

So the dependence relation is:  $-2\mathbf{b}_1 - 1\mathbf{b}_2 + 1\mathbf{b}_3 = \mathbf{0}$

$$\text{Null}(B) = \text{span} \left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$

## Verify with Our Example (continued)

$$AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

**Check:** Does  $\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \in \text{Null}(AB)$ ?

$$AB \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 - 1 + 5 \\ -2 + 0 + 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Yes! ✓

**Find  $\text{Null}(AB)$  completely:**

Column 3 of  $AB$  equals  $2 \times$  column 1 plus  $1 \times$  column 2:

$$\text{Null}(AB) = \text{span} \left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$



# The Big Question

We've seen:

- $\text{Col}(AB) \subseteq \text{Col}(A)$  (always)
- $\text{Null}(B) \subseteq \text{Null}(AB)$  (always)

**But:** In our example, we had **equality** in both cases!

**When is  $\text{Null}(AB) = \text{Null}(B)$  exactly?**

**Meaning:** When does multiplication by  $A$  **not create new dependencies?**

## Theorem 3: The Equivalence

### Theorem

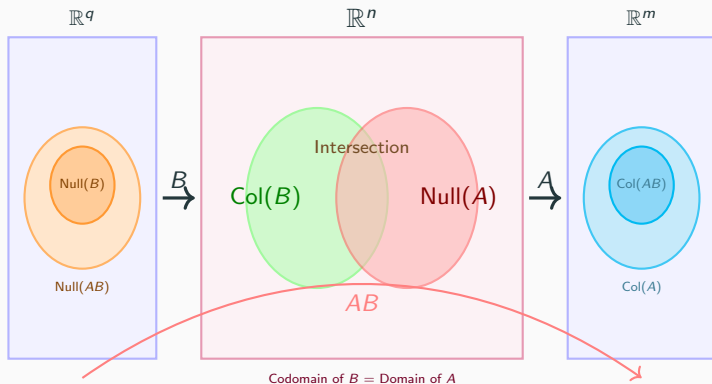
$$\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \iff \text{Null}(AB) = \text{Null}(B)$$

**Left side:** Among all vectors  $B$  can produce, none (except  $\mathbf{0}$ ) lands in  $\text{Null}(A)$ .

**Right side:** All dependencies in  $AB$  were already in  $B$  (no new ones created).

These are **two perspectives on the same phenomenon!**

# Visual: The Equivalence



$$\text{Null}(AB) = \text{Null}(B) \iff \text{Col}(B) \cap \text{Null}(A) = \{\mathbf{0}\}$$

## Proof Direction (a): $\Rightarrow$

**Assume:**  $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

**Goal:** Show  $\text{Null}(AB) = \text{Null}(B)$

We already know  $\text{Null}(B) \subseteq \text{Null}(AB)$  (Theorem 2).

To show equality, prove the reverse inclusion.

Take any  $\mathbf{x} \in \text{Null}(AB)$ . Then  $(AB)\mathbf{x} = \mathbf{0}$ , so:

$$A(B\mathbf{x}) = \mathbf{0}$$

This means  $B\mathbf{x} \in \text{Null}(A)$ .

## Proof Direction (a): $\Rightarrow$ (continued)

We have:  $B\mathbf{x} \in \text{Null}(A)$ .

But also:  $B\mathbf{x} \in \text{Col}(B)$  (by definition of column space).

Therefore:  $B\mathbf{x} \in \text{Null}(A) \cap \text{Col}(B)$ .

By assumption,  $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$ , so:

$$B\mathbf{x} = \mathbf{0}$$

This means  $\mathbf{x} \in \text{Null}(B)$ .

We've shown  $\text{Null}(AB) \subseteq \text{Null}(B)$ . Combined with Theorem 2:

$$\text{Null}(AB) = \text{Null}(B)$$



## Proof Direction (b): $\Leftarrow$

**Assume:**  $\text{Null}(AB) = \text{Null}(B)$

**Goal:** Show  $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

Suppose  $\mathbf{v} \in \text{Null}(A) \cap \text{Col}(B)$ .

Since  $\mathbf{v} \in \text{Null}(A)$ :  $A\mathbf{v} = \mathbf{0}$ .

Since  $\mathbf{v} \in \text{Col}(B)$ :  $\mathbf{v} = B\mathbf{x}$  for some  $\mathbf{x}$ .

Combining:

$$A(B\mathbf{x}) = A\mathbf{v} = \mathbf{0}$$

So  $(AB)\mathbf{x} = \mathbf{0}$ , meaning  $\mathbf{x} \in \text{Null}(AB)$ .

## Proof Direction (b): $\Leftarrow$ (continued)

We have:  $\mathbf{x} \in \text{Null}(AB)$ .

By hypothesis,  $\text{Null}(AB) = \text{Null}(B)$ , so:

$$\mathbf{x} \in \text{Null}(B)$$

Therefore:

$$B\mathbf{x} = \mathbf{0}$$

But we had  $\mathbf{v} = B\mathbf{x}$ , so:

$$\mathbf{v} = \mathbf{0}$$

We've shown: any  $\mathbf{v} \in \text{Null}(A) \cap \text{Col}(B)$  must be  $\mathbf{0}$ .

Therefore:  $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$



## Summary: Subspaces of Matrix Products

| Theorem                                                                                                         | Meaning                                   |
|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| $\text{Col}(AB) \subseteq \text{Col}(A)$                                                                        | Columns of $AB$ built from $A$ 's columns |
| $\text{Null}(B) \subseteq \text{Null}(AB)$                                                                      | Dependencies in $B$ persist in $AB$       |
| $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$<br>$\Leftrightarrow$<br>$\text{Null}(AB) = \text{Null}(B)$ | No new dependencies created               |

**Key insight:** Matrix multiplication has predictable effects on subspaces.

**Next:** We need a new tool to reveal deeper relationships — **transpose**.



## **Transpose: Flipping the Production Table**

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# What Is Transpose? Flipping Perspective

In our ingredient table framework:

- **Rows** = raw materials (inputs)
- **Columns** = products (outputs)
- **Entry  $A_{ij}$**  = amount of material  $i$  needed for product  $j$

**Original question:** Given a product, how much of each material do we need?

**Flipped question:** Given a material, which products use it?

This perspective flip is called **transpose**.

# Concrete Example: Coffee Shop Ingredient Table

Original table  $A$  (materials  $\rightarrow$  products):

|                                                                                   |  |  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  | 0                                                                                 | 0                                                                                 | 2                                                                                 |
|  | 0                                                                                 | 0                                                                                 | 1                                                                                 |
|  | 0                                                                                 | 2                                                                                 | 4                                                                                 |
|  | 1                                                                                 | 0                                                                                 | 1                                                                                 |

$$= \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

Reading:

- Column  : needs  $(0 \text{ , 0 \text{ , 0 \text{ , 1 \text{ $ )
- Row  : distributed to products as  $(0, 0, 2)$

# Flipped Table: Transpose $A^T$

Flipped table  $A^T$  (products  $\rightarrow$  materials):

|                                                                                   |  |  |  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  | 0                                                                                 | 0                                                                                 | 0                                                                                 | 1                                                                                 |
|  | 0                                                                                 | 0                                                                                 | 2                                                                                 | 0                                                                                 |
|  | 2                                                                                 | 1                                                                                 | 4                                                                                 | 1                                                                                 |

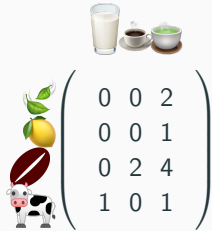
$$= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 4 & 1 \end{pmatrix}$$

Reading:

- Row  : contains amounts  $(0 \text{  , 0 \text{  , 0 \text{  , 1 \text{  )}$
- Column  : contributes to products as  $(0, 0, 2)$

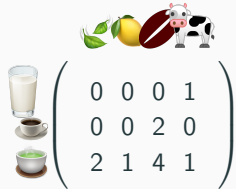
**Notice:** Rows of  $A$  become columns of  $A^T$ , and vice versa!

# Visual: Flipping the Table


$$\begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$

$A$  (materials = rows)




$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 2 & 1 & 4 & 1 \end{pmatrix}$$

$A^T$  (products = rows)

**Key:** The rows and columns swap roles!

# Formal Definition

## Definition (Transpose)

For an  $m \times n$  matrix  $A$ , its **transpose**  $A^T$  is the  $n \times m$  matrix:

$$(A^T)_{ij} = A_{ji}$$

Rows of  $A$  become columns of  $A^T$ , and columns of  $A$  become rows of  $A^T$ .

**Example:**

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Row 1 of  $A$ :  $(1, 2, 3)$  becomes column 1 of  $A^T$ :  $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

# Property 1: Transpose of Product

## Proposition

$$(AB)^T = B^T A^T$$

The transpose of a product reverses the order!

**Proof** (entry-by-entry):

Entry  $(i, j)$  of  $(AB)^T$ :

$$(AB)_{ij}^T = (AB)_{ji} = \sum_k A_{jk} B_{ki}$$

Entry  $(i, j)$  of  $B^T A^T$ :

$$(B^T A^T)_{ij} = \sum_k (B^T)_{ik} (A^T)_{kj} = \sum_k B_{ki} A_{jk}$$

Since multiplication of numbers commutes:  $\sum_k A_{jk} B_{ki} = \sum_k B_{ki} A_{jk}$   $\square$

# Why the Order Reverses: Production View

In a two-stage production  $A \cdot B$ :

- $B$ : raw materials  $\rightarrow$  semi-products
- $A$ : semi-products  $\rightarrow$  final products

**Forward direction** ( $AB$ ): raw  $\xrightarrow{B}$  semi  $\xrightarrow{A}$  final

**Flipped perspective** (transpose):

- $A^T$ : final product  $\rightarrow$  which semi-products it uses
- $B^T$ : semi-product  $\rightarrow$  which raw materials it needs

**Backward tracing**: final  $\xrightarrow{A^T}$  semi  $\xrightarrow{B^T}$  raw

Hence  $(AB)^T = B^T A^T$  — the order flips!



## Property 2: Transpose of Inverse

### Proposition

If  $A$  is invertible:

$$(A^{-1})^T = (A^T)^{-1}$$

Transpose and inverse commute.

### Proof:

Start from  $A \cdot A^{-1} = I$ , transpose both sides:

$$(A \cdot A^{-1})^T = I^T = I$$

Using Property 1:

$$(A^{-1})^T \cdot A^T = I$$

Similarly,  $(A^{-1} \cdot A)^T = I$  gives  $A^T \cdot (A^{-1})^T = I$ .

Therefore  $(A^{-1})^T$  is the inverse of  $A^T$ .



## Property 3: Transpose is Linear

### Proposition

$$(A + B)^T = A^T + B^T$$

Transpose distributes over addition.

**Proof** (entry-by-entry):

$$\begin{aligned}(A + B)_{ij}^T &= (A + B)_{ji} \\ &= A_{ji} + B_{ji} \\ &= (A^T)_{ij} + (B^T)_{ij} \\ &= (A^T + B^T)_{ij}\end{aligned}$$

Therefore  $(A + B)^T = A^T + B^T$ . □

# Symmetric Matrices

## Definition (Symmetric Matrix)

A matrix  $S$  is **symmetric** if  $S^T = S$ .

The ingredient table looks the same when flipped!

**Example:**

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$$

Flipping rows  $\leftrightarrow$  columns gives the same matrix.

**Notice:**  $S_{ij} = S_{ji}$  for all  $i, j$  (mirror symmetry across diagonal).

# Theorem: $AA^T$ and $A^T A$ Are Always Symmetric

## Theorem

For any matrix  $A$ :

1.  $AA^T$  is symmetric
2.  $A^T A$  is symmetric

## Proof:

Using  $(AB)^T = B^T A^T$ :

$$(AA^T)^T = (A^T)^T A^T = AA^T$$

So  $AA^T$  is symmetric.

Similarly:

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

So  $A^T A$  is symmetric.



## Verify with Concrete Matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Compute  $AA^T$ :

$$AA^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

Check:  $(AA^T)^T = AA^T \checkmark$  (symmetric!)

## Verify with Concrete Matrix (continued)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Compute  $A^T A$ :

$$A^T A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 35 & 44 \\ 44 & 56 \end{pmatrix}$$

Check:  $(A^T A)^T = A^T A \checkmark$  (symmetric!)

# Change of Perspective

What we've learned so far:

- **Why** transpose exists: flipping the ingredient table perspective
- **Why**  $(AB)^T = B^T A^T$ : backward tracing reverses order
- **Why**  $AA^T$  is symmetric: algebraic consequence of transpose properties

## From This Point Forward

We **forget about coffee shops** and work with abstract matrices  $A$ ,  $B$ ,  $C$ .

The key properties:  $(AB)^T = B^T A^T$ , symmetry of  $AA^T$  and  $A^T A$ .

These apply to **all matrices**, and we'll use them to discover deeper geometric relationships.

**Next:** The four fundamental subspaces and their beautiful orthogonality.

# Four Fundamental Subspaces and Orthogonality

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# The Four Fundamental Subspaces

For any  $m \times n$  matrix  $A$ , there are **four fundamental subspaces**:

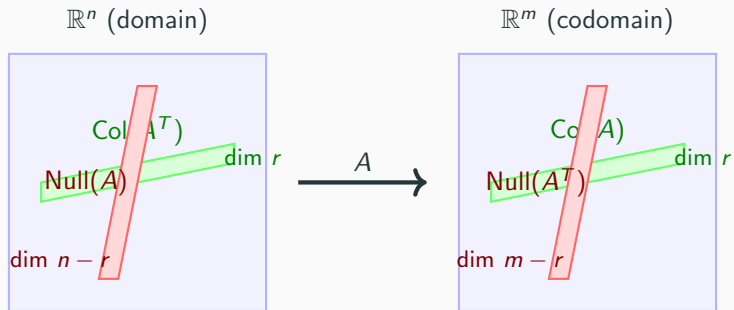
| Subspace           | Definition                                    | Lives in       | Dimension |
|--------------------|-----------------------------------------------|----------------|-----------|
| $\text{Col}(A)$    | $\{A\mathbf{x}\}$                             | $\mathbb{R}^m$ | $r$       |
| $\text{Null}(A)$   | $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$   | $\mathbb{R}^n$ | $n - r$   |
| $\text{Col}(A^T)$  | $\{A^T\mathbf{y}\}$                           | $\mathbb{R}^n$ | $r$       |
| $\text{Null}(A^T)$ | $\{\mathbf{y} : A^T\mathbf{y} = \mathbf{0}\}$ | $\mathbb{R}^m$ | $m - r$   |

where  $r = \text{rank}(A)$ .

## Key observations:

- Two subspaces in  $\mathbb{R}^n$  (domain):  $\text{Null}(A)$  and  $\text{Col}(A^T)$
- Two subspaces in  $\mathbb{R}^m$  (codomain):  $\text{Col}(A)$  and  $\text{Null}(A^T)$

## Visual: Two Pairs of Subspaces



**Today's discovery:** These pairs are **orthogonal** to each other!

## Theorem 4: $\text{Col}(A^T) \perp \text{Null}(A)$

### Theorem

Every vector in  $\text{Col}(A^T)$  is orthogonal to every vector in  $\text{Null}(A)$ .

Precisely: For  $\mathbf{x} \in \text{Null}(A)$  and  $\mathbf{y} \in \text{Col}(A^T)$ ,

$$\mathbf{y}^T \mathbf{x} = 0$$

**Meaning:** The row space and null space are perpendicular!

## Proof (Inner Product View)

Let  $\mathbf{x} \in \text{Null}(A)$  and  $\mathbf{y} \in \text{Col}(A^T)$ .

By definition:

- $\mathbf{x} \in \text{Null}(A)$  means  $A\mathbf{x} = \mathbf{0}$
- $\mathbf{y} \in \text{Col}(A^T)$  means  $\mathbf{y} = A^T\mathbf{z}$  for some  $\mathbf{z}$

Compute the inner product:

$$\begin{aligned}\mathbf{y}^T \mathbf{x} &= (A^T \mathbf{z})^T \mathbf{x} \\ &= \mathbf{z}^T (A^T)^T \mathbf{x} \\ &= \mathbf{z}^T A \mathbf{x}\end{aligned}$$

Since  $\mathbf{x} \in \text{Null}(A)$ , we have  $A\mathbf{x} = \mathbf{0}$ :

$$\mathbf{z}^T A \mathbf{x} = \mathbf{z}^T \mathbf{0} = 0$$



## Proof (Geometric View: Rows as Normal Vectors)

The equation  $A\mathbf{x} = \mathbf{0}$  can be written row-by-row:

$$\begin{pmatrix} -\mathbf{r}_1^T - \\ -\mathbf{r}_2^T - \\ \vdots \\ -\mathbf{r}_m^T - \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{r}_1^T \mathbf{x} \\ \mathbf{r}_2^T \mathbf{x} \\ \vdots \\ \mathbf{r}_m^T \mathbf{x} \end{pmatrix} = \mathbf{0}$$

This means:

$$\mathbf{r}_1^T \mathbf{x} = 0, \quad \mathbf{r}_2^T \mathbf{x} = 0, \quad \dots, \quad \mathbf{r}_m^T \mathbf{x} = 0$$

**Interpretation:** Each row  $\mathbf{r}_i$  is **perpendicular** to  $\mathbf{x}$ .

## Proof (Geometric View: continued)

We have: Each row  $\mathbf{r}_i$  of  $A$  is perpendicular to every  $\mathbf{x} \in \text{Null}(A)$ .

**But:** The rows of  $A$  are exactly the **columns of  $A^T$** !

$$A^T = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{r}_1 & \mathbf{r}_2 & \cdots & \mathbf{r}_m \\ | & | & \cdots & | \end{pmatrix}$$

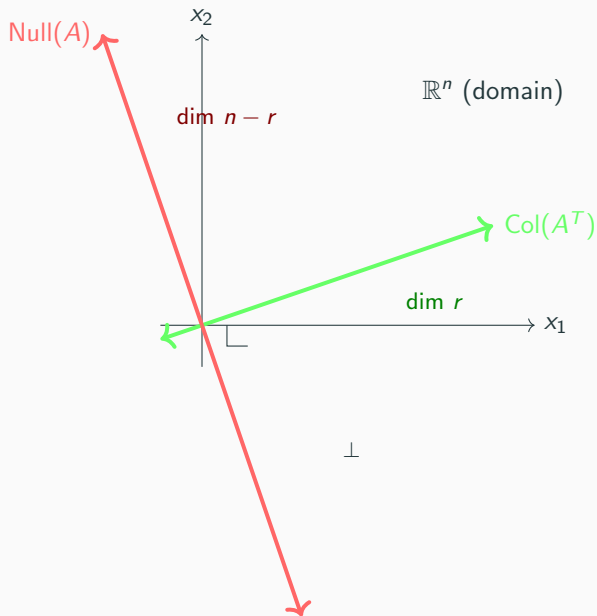
Therefore:

$$\text{Col}(A^T) = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m\}$$

consists of vectors perpendicular to  $\text{Null}(A)$ .



## Visual: Orthogonality in $\mathbb{R}^n$



## Concrete Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

**Find Null( $A$ ):**

$A\mathbf{x} = \mathbf{0}$  gives:  $x_1 + 2x_2 + 3x_3 = 0$  (second row is  $2\times$  first)

Basis for Null( $A$ ):

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

**Find Col( $A^T$ ):**

$$A^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$$



## Verify Orthogonality

$$\text{Col}(A^T) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}, \quad \text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Check orthogonality:

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}^T \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = 1(-2) + 2(1) + 3(0) = 0 \quad \checkmark$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}^T \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} = 1(-3) + 2(0) + 3(1) = 0 \quad \checkmark$$

Indeed,  $\text{Col}(A^T) \perp \text{Null}(A)$ !

## Theorem 5: $\text{Col}(A) \perp \text{Null}(A^T)$

### Theorem

Every vector in  $\text{Col}(A)$  is orthogonal to every vector in  $\text{Null}(A^T)$ .

Precisely: For  $\mathbf{x} \in \text{Col}(A)$  and  $\mathbf{y} \in \text{Null}(A^T)$ ,

$$\mathbf{y}^T \mathbf{x} = 0$$

**Meaning:** The column space and left null space are perpendicular!

# Proof

Let  $\mathbf{y} \in \text{Null}(A^T)$  and  $\mathbf{x} \in \text{Col}(A)$ .

By definition:

- $\mathbf{y} \in \text{Null}(A^T)$  means  $A^T \mathbf{y} = \mathbf{0}$
- $\mathbf{x} \in \text{Col}(A)$  means  $\mathbf{x} = A\mathbf{z}$  for some  $\mathbf{z}$

Compute the inner product:

$$\begin{aligned}\mathbf{y}^T \mathbf{x} &= \mathbf{y}^T (A\mathbf{z}) \\ &= (A^T \mathbf{y})^T \mathbf{z}\end{aligned}$$

Since  $A^T \mathbf{y} = \mathbf{0}$ :

$$(A^T \mathbf{y})^T \mathbf{z} = \mathbf{0}^T \mathbf{z} = 0$$



# Symmetry of the Theorems

## Beautiful duality:

- **Theorem 4:**  $\text{Col}(A^T) \perp \text{Null}(A)$  (both in  $\mathbb{R}^n$ )
- **Theorem 5:**  $\text{Col}(A) \perp \text{Null}(A^T)$  (both in  $\mathbb{R}^m$ )

These are **not two separate facts** — they're the **same theorem!**

**Why?** Apply Theorem 4 to the matrix  $A^T$  instead of  $A$ :

$$\text{Col}((A^T)^T) \perp \text{Null}(A^T) \quad \Rightarrow \quad \text{Col}(A) \perp \text{Null}(A^T)$$

which is exactly Theorem 5!

**Deep insight:** Transpose reveals a fundamental symmetry in linear algebra.

## Concrete Example: The Dual Case

Using the same matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

**Find  $\text{Col}(A)$ :**

$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix} \right\} = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$$

**Find  $\text{Null}(A^T)$ :**

$$A^T \mathbf{y} = \mathbf{0} \text{ gives: } \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{0}$$

All equations reduce to:  $y_1 + 2y_2 = 0$

$$\text{Null}(A^T) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$$

## Concrete Example: Verify Orthogonality

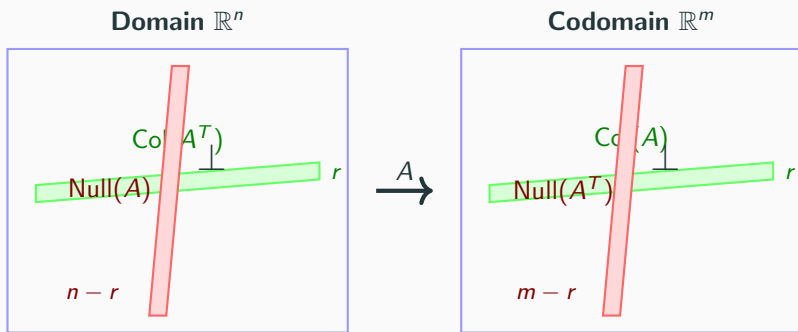
$$\text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}, \quad \text{Null}(A^T) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$$

Check orthogonality:

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 2 \end{pmatrix} = (-2)(1) + (1)(2) = -2 + 2 = 0 \quad \checkmark$$

Indeed,  $\text{Col}(A) \perp \text{Null}(A^T)$ !

# The Big Picture: Four Subspaces



$$\dim \text{Col}(A^T) + \dim \text{Null}(A) = r + (n - r) = n = \dim \text{Col}(A) + \dim \text{Null}(A^T) = r + (m - r) = m$$

**Key:** The orthogonal pairs completely fill their spaces!

# Dimension Relationships

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## Theorem 6: Row Rank Equals Column Rank

### Theorem

$$\text{rank}(A) = \text{rank}(A^T)$$

The dimension of  $\text{Col}(A)$  equals the dimension of  $\text{Col}(A^T)$ .

In other words: **Column rank = Row rank**.

**Proof idea** (using cross-filling from Lecture 3):

Any matrix can be decomposed:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where  $U$  is  $m \times r$  and  $V$  is  $r \times n$ .

## Proof of Theorem 6 (continued)

$$A = UV$$

where:

- $U$  is  $m \times r$  with  $r$  linearly independent columns
- $V$  is  $r \times n$  with  $r$  linearly independent rows

**Column rank:**

$$\text{Col}(A) = \text{Col}(UV) = \text{Col}(U)$$

So  $\dim(\text{Col}(A)) = r$ .

**Row rank:**

The rows of  $A = UV$  are linear combinations of the rows of  $V$ .

Since  $V$  has  $r$  linearly independent rows,  $\dim(\text{Col}(A^T)) = r$ .

Therefore  $\text{rank}(A) = \text{rank}(A^T) = r$ .



# Dimension Formulas

From the rank-nullity theorem (Lecture 5) and orthogonality:

## Proposition

For an  $m \times n$  matrix  $A$  with  $\text{rank}(A) = r$ :

In  $\mathbb{R}^n$  (domain):

$$\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = r + (n - r) = n$$

In  $\mathbb{R}^m$  (codomain):

$$\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = r + (m - r) = m$$

**Meaning:** The orthogonal pairs together **span their entire spaces**.

## Summary

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# What We Achieved Today

## 1. Subspaces of Matrix Products:

- $\text{Col}(AB) \subseteq \text{Col}(A)$  — columns of  $AB$  built from  $A$ 's columns
- $\text{Null}(B) \subseteq \text{Null}(AB)$  — dependencies persist
- $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \Leftrightarrow \text{Null}(AB) = \text{Null}(B)$  — no new dependencies

## 2. Transpose:

- Flipping the ingredient table perspective
- $(AB)^T = B^T A^T$  — order reverses
- $AA^T$  and  $A^T A$  always symmetric

# What We Achieved Today (continued)

## 3. Four Fundamental Subspaces:

- $\text{Col}(A^T) \perp \text{Null}(A)$  in  $\mathbb{R}^n$
- $\text{Col}(A) \perp \text{Null}(A^T)$  in  $\mathbb{R}^m$
- These are dual statements — same theorem applied to  $A$  and  $A^T$

## 4. Dimension Relationships:

- $\text{rank}(A) = \text{rank}(A^T)$  — column rank equals row rank
- $\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = n$
- $\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = m$

**Key insight:** The four fundamental subspaces form two orthogonal pairs that completely describe the geometry of a matrix.

## Lecture 7: Linear Transformations and Projections

Now that we understand the four fundamental subspaces and their orthogonality, we can study:

- What are linear transformations?
- What are projection operators? ( $P^2 = P$ )
- How do projections relate to the four subspaces?
- The geometric meaning of  $\text{Col}(P)$  and  $\text{Null}(P)$

**Goal:** Understand how to decompose vectors into orthogonal components.